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Question 2 · parts 2-1 to 2-4

2-1. Lecture 10 p. 9: $n_s = 120f/p$. For the same pole count, $n'/n = f'/f$. Field current changes flux and spatial phase sequence changes direction. **Choice A.**

2-2. Lecture 10 p. 8 constructs the rotating field. Write $B(t) = B_0[\cos(\omega t)\hat{x} + \sin(\omega t)\hat{y}]$. Interchanging two spatial phase assignments gives $B_{rev}(t) = B_0[\cos(\omega t)\hat{x} - \sin(\omega t)\hat{y}]$. The field rotates in reverse. **Choice C.**

2-3. Lecture 10 p. 14: $|E_A| = 4.44fN\Phi$. At fixed f, field current changes Φ; in a linear magnetic circuit $\Phi = k_f I_f$, hence $|E'|/|E| = I'_f/I_f$. **Intended choice B at fixed frequency.**

Source ambiguity: fixed frequency is not printed. Changing f with unchanged flux also changes internal voltage, so A is possible under a literal reading. Saturation prevents assuming exact field-current proportionality in general.

2-4. Lecture 10 p. 9: $\omega_m = (2\pi/60)(120f/p) = 4\pi f/p$. The pole count is unchanged; $\omega_{m2}/\omega_{m1} = 60/50 = 1.2$. Equal output horsepower changes torque, not synchronous speed. **$\omega_{m1} < \omega_{m2}$; choice B.**

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Question 2 · parts 2-5 to 2-7

2-5. Lecture 4 p. 26: $S_{two} = V_1 I_1 = V_2 I_2$. Connect winding 1 as common winding and winding 2 in series-aiding. Output voltage is $V_1 + V_2$; series current is limited to I_2. Therefore $S_{auto} = (V_1 + V_2)I_2 = (1 + V_1/V_2)S_{two} > S_{two}$, subject to insulation and connection ratings. **Choice A, True.**

2-6. Lecture 10 pp. 9, 14: $f = pn/120$ and $|E_A| = 4.44 f N \Phi$. A fixed RL series load has $pf = R/\sqrt{[R^2 + (2\pi f L)^2]}$. At fixed flux, increasing n increases f and $|E_A|$, while power factor decreases. **Choice A under fixed excitation.** The question does not specify a regulator; an isolated generator is assumed, not an infinite-bus-constrained machine.

2-7. Use spatial MMF superposition (Lecture 10 pp. 8, 14). All three currents are $i(t) = I_{pk}\cos(120\pi t + \alpha)$. With winding axes 0, 2π/3, 4π/3, the fundamental is

$$F_1(\theta,t) = N_1 i(t)[\cos\theta + \cos(\theta - 2\pi/3) + \cos(\theta - 4\pi/3)] = 0.$$

The cosine coefficients sum to $1 - 1/2 - 1/2 = 0$, and the sine coefficients sum to $0 + \sqrt{3}/2 - \sqrt{3}/2 = 0$. A third spatial harmonic instead gives

$$F_3(\theta,t) = 3N_3 I_{pk}\cos(3\theta)\cos(120\pi t + \alpha).$$

Its nodes are fixed and its amplitude pulsates. **C can occur with spatial harmonics; the ideal fundamental is zero.** Under the lecture’s purely sinusoidal winding model none of the printed choices describes a nonzero fundamental. Equal phase currents require independent supplies or a neutral/return path; a three-wire star circuit cannot support them. The unspecified RMS/peak current convention does not affect cancellation.

Question 3 · loaded Y-connected generator

(a) Lecture 10 p. 18: $E_A = V_\varphi + jX_sI_A$, with current flowing out of the generator; $V_\varphi = Z_LI_A$. The single-phase circuit below has a common neutral return. The source is in series with $Z_s = j2\ \Omega$ and the phase load $Z_L = 10+j10\ \Omega$. Total series impedance is **$10+j12\ \Omega$**. Both Y connections give $I_L = I_A$.

(b) Lecture 10 p. 14 gives $|E_A|$ proportional to $f\Phi$. At no load the phase current is zero, so $|E_{A0}| = V_{LL0}/\sqrt{3} = 100/\sqrt{3} = 57.7350$ V.

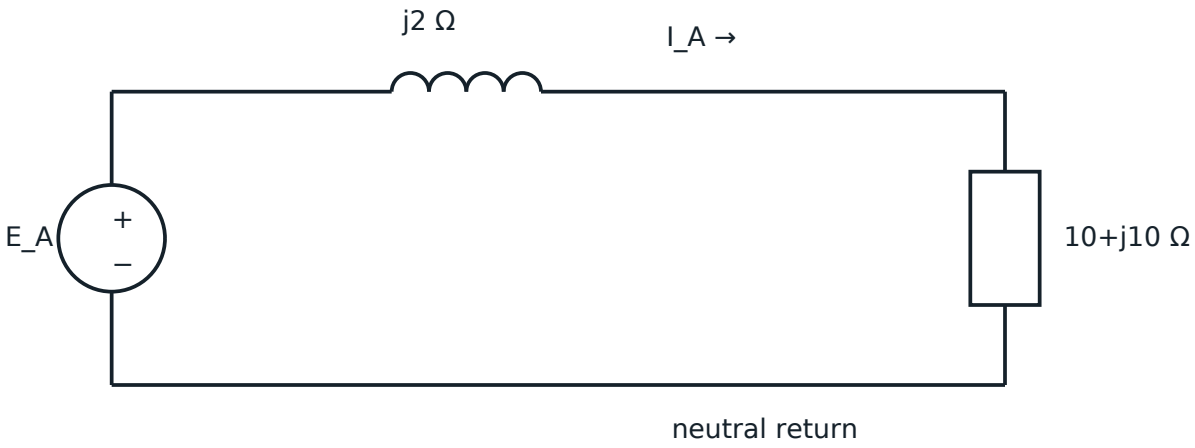
At unchanged speed and assuming an unsaturated linear magnetic circuit, $|E_{A1}| = (20/10)(100/\sqrt{3}) = 115.4701$ V.

From the series circuit, $V_\varphi = E_{A1}Z_L/(Z_L + jX_s) = (200/\sqrt{3})(10+j10)/(10+j12)$ V.

$|V_\varphi| = (200/\sqrt{3})\sqrt{(10^2+10^2)}/\sqrt{(10^2+12^2)} = 104.5417$ V.

$V_{LL} = \sqrt{3}(104.5417) = 200\sqrt{(200/244)} = 181.0715$ V RMS.

Source correction: “primer mover” is printed. The printed j2 Ω is impedance; $X_s = 2\ \Omega$. Doubling field current does not guarantee doubled flux with saturation (Lecture 10 p. 15). Without the assumed linear magnetization and unchanged reactance, no unique numerical loaded voltage follows from the supplied data.



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Question 4 · turbine-driven generator

(a) Lecture 10 p. 9 gives $n_s = 120f/p$; mechanical power is $P = T\omega_m$. Substituting $f = 60$ Hz and $p = 2$ gives $n_s = 3600$ rpm and $\omega_m = 2\pi(3600)/60 = 120\pi$ rad/s. With all losses neglected, $P_{mech} = 1.5\times10^6$ W. **$T = 1.5\times10^6/(120\pi) = 3978.87$ N m.**

(b) Lecture 10 p. 24: $P = 3|E_A|V_\varphi \sin\delta/X_s$. For Y connection, $V_\varphi = 1732/\sqrt{3} = 999.970666$ V. Therefore

$|E_A| = PX_s/(3V_\varphi \sin\delta) = (1.5\times10^6)(1)/[3(1732/\sqrt{3})\sin 30°]$ = **1000.029335 V RMS per phase.**

(c) Lecture 10 pp. 18, 20 give $I_A = (E_A - V_\varphi)/(jX_s)$ and $S_{load} = 3V_\varphi I_A^*$. Choose V_φ as the zero-angle reference and use the result of (b):

$I_A = [1000.029335(\cos 30° + j \sin 30°) - 999.970666]/j = 500.014667 + j133.919858$ A.

Current angle = arctan(133.919858/500.014667) = +14.993727°. The load current leads its voltage, and its received reactive power is negative.

pf = 500.014667/√(500.014667² + 133.919858²) = 0.965954 leading.

Source convention: j1 Ω is the synchronous impedance, so $X_s = 1$ Ω. The printed 1.732 kV is retained exactly rather than replaced by 1000√3 V.

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Question 5 · delta-connected motor

(a) Reverse the generator-current convention of Lecture 10 pp. 18, 24. With motor current inward and δ a positive lag, $P_{dev} = 3V_{\varphi}|E_A|\sin\delta/X_s$; $I_{\varphi} = [V_{\varphi} - |E_A|\angle(-\delta)]/(jX_s)$. Delta gives $V_{\varphi} = 600$ V and $I_L = \sqrt{3}|I_{\varphi}|$.

$|E_{A,a}| = 40000(4)/[3(600)\sin 60°] = 102.640048$ V.

$I_{\varphi,a} = [600 - 102.640048(0.5 - j0.8660254)]/(j4) = 22.222222 - j137.169994$ A. Its magnitude is $\sqrt{(22.222222^2 + 137.169994^2)} = 138.958391$ A. **$I_{L,a} = \sqrt{3}(138.958391) = 240.682993$ A RMS.**

(b) Differentiate P_{dev} with respect to δ: $dP/d\delta = (3V_{\varphi}|E_A|/X_s)\cos\delta$. Its positive maximum is at δ = 90°, where the second derivative is negative. Linear excitation gives $|E_{A,b}| = 1.2(102.640048) = 123.168057$ V.

$I_{\varphi,b} = [600 - 123.168057\angle(-90°)]/(j4) = 30.792014 - j150$ A. $|I_{\varphi,b}| = \sqrt{(30.792014^2 + 150^2)} = 153.127882$ A.

New torque angle = 90° lag (signed internal-voltage angle −90°); $I_{L,b} = \sqrt{3}(153.127882) = 265.225271$ A RMS.

(c) Lecture 10 p. 20: $S_{in} = 3V_{\varphi}I_{\varphi}^*$. Continuing (b), $S_{in,b} = 3(600)(30.792014 + j150) = 55425.626 + j270000$ VA. Positive Q means motor absorption. **$Q_{in,b} = 270.000$ kvar and winding current $= 265.225271/\sqrt{3} = 153.127882$ A RMS.**

Source ambiguities: “negligible armature” is incomplete; negligible armature resistance is assumed. “Adjust the prime mover” is interpreted as adjusting mechanical shaft loading. Exact maximum power is the ideal stability boundary. Field-current scaling assumes no saturation. Part (c) does not identify its operating condition: if it instead refers to (a), $Q_{in,a} = 3(600)(137.169994) = 246.905989$ kvar and winding current = 138.958391 A. The printed data alone do not resolve that wording.